

UMA005: INTRODUCTORY MATHEMATICS-I

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective is to develop the basics of computing skills and application of quantitative and statistical operations required for biological studies.

Syllabus

Algebra: Complex numbers, Solution of quadratic equations, Permutations and combinations, Binomial theorem for positive/negative index and its simple applications, Arithmetic and geometric progression

Trigonometry: Review of trigonometric functions, Sum and product formulae for trigonometric functions, Trigonometric equations and sum - to - product formulae for trigonometric functions, Identities related to double angle formulae

Determinants and Matrices: Matrices, Operations on matrices, Determinants and its properties, Singular and non-singular matrices, Adjoint and inverse of a matrix and its properties, Solution of system of linear equations using Cramer's rule and matrix method

Coordinate Geometry: Rectangular coordinate system, Straight lines, Circles (in standard form only)

Statistics: Measure of dispersion: mean deviation, Variance and standard deviation of grouped/ungrouped data. Correlation and regression

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. acquire knowledge of basic algebra, trigonometry, matrices, coordinate geometry etc.
2. apply these concepts to solve complex mathematical problems
3. analyse the data of any experiment statistically to extract meaningful result
4. tackle any mathematical challenge that usually occurs during their biological studies.

Text Books

1. Mathematics, A Text books (Parts I & II), NCERT, New Delhi (2011).
2. Kreyszig, Erwin, Advanced Engineering Mathematics, John Wiley, (1999).

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Reference Books

1. Krishnamurthy V.K., Mainra V.P. and Arora J.L. An introduction to Linear Algebra, Associated East West Press (2007).
2. Loney, S. L., The elements of Coordinate Geometry, Michigan Historical Reprint series, (2012)
3. Meyer, P. L., Introductory Probability and Statistical Applications, Addison Wesley (1970).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/tutorials/quizzes)	30

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